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package rooms;

public class Room {

    private int roomNumber;
    private String roomType; // single, double, suite
    private boolean occupied;

    private static int count = 0; // to keep track of created objects

    public Room() {
        this(1, "single", false);
    }

    public Room(int rn, String givenRoomType, boolean occ) {
        setRoomNumber(rn);
        setRoomType(givenRoomType);
        occupied = occ;
        count++;
    }

    public int getRoomNumber() {
        return roomNumber;
    }

    //this method does not touch the instance variables
    // it is better if it stays as static
    public static boolean isValidRoomNo(int x){
        if(x>0 && x<1000){
            int val=x/100;
            if(val*100+20>x){
                return true;
            }
        }
        return false;
    }

    // this method should be only callable by the class
    private void setRoomNumber(int rn) {
        if(isValidRoomNo(rn)){
            roomNumber = rn;
        }else{
            System.out.println("Error in setting room number");
            roomNumber = 999;
        }
    }

    public boolean isOccupied() {
        return occupied;
    }

    public void checkin() {
        if (!isOccupied()) {
            occupied = true;
        }
    }

    public void checkout() {
        occupied = false;
    }
}

```

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    }

    public void setRoomType(String givenRoom) {
        if (givenRoom.equals("single") ||
            givenRoom.equals("double") ||
            givenRoom.equals("suite")) {
            roomType = givenRoom;
        } else {
            System.out.println("Invalid room type entered");
        }
    }

    public String getRoomType() {
        return roomType;
    }

    public String toString() {
        String result = "number " + roomNumber;
        if (isOccupied()){
            result += " is occupied";
        }else{
            result += " is vacant";
        }
        return result;
    }

    public void showRoom() {
        String result = "[ ]"; // assume the room is vacant
        if (isOccupied())
            result = "[x]"; // if occupied change the output
        System.out.print(result);
    }

    public static int getCount() { // since count is static this method should be
static
        return count;
    }

} // end of class

```